

Sleipner CO₂ Storage Project

30 Years of Safe Geological CO₂ Storage — The Benchmark for Permanence

QUICK FACTS

- Sector:** Geological Storage / Offshore Natural Gas Processing
- Location:** North Sea, Norway (Sleipner West field)
- Technology:** CO₂ injection into Utsira saline aquifer; amine capture on production platform

1. Background: Why Sleipner Exists

The Sleipner gas field is a natural gas production platform operated by Equinor (formerly Statoil) in the North Sea, approximately 250 km west of Stavanger, Norway. The natural gas produced at Sleipner contains approximately 9% CO₂ — far above the 2.5% maximum permitted by European pipeline specifications.

To sell the gas, the CO₂ must be removed. The traditional solution would be to vent the stripped CO₂ to the atmosphere — standard industry practice globally. But in 1991, Norway introduced an offshore CO₂ tax of approximately USD 50 per tonne, making venting economically unattractive.

Equinor engineers developed an alternative: inject the removed CO₂ into the Utsira sandstone formation — a deep saline aquifer 800–1,000 metres below the seabed — directly beneath the production platform. The economics made sense, the geology was promising, and the injection began on 15 August 1996.

Nearly 30 years later, Sleipner remains the world's longest-running continuous CO₂ geological storage operation and the most thoroughly studied CCS site on Earth.

Operator	Equinor (formerly Statoil)
Start date	15 August 1996
Storage formation	Utsira sandstone, 800–1,000 m below seabed
Annual injection rate	~0.85–1.1 million tonnes CO ₂ /year
Total injected (to 2024)	~25 million tonnes CO ₂
Storage temperature	~37°C (supercritical conditions)
Storage pressure	~80–100 bar
Estimated storage capacity	600 billion tonnes CO ₂ (Utsira formation, full extent)

Detected leakage	Zero — in 28+ years of continuous monitoring
Economic driver	Norwegian offshore CO ₂ tax (~\$50–80/tonne, adjusted over years)

2. Geology: Why the Utsira Formation Works

The Utsira sandstone is an exceptionally good CO₂ storage formation. Its properties explain why Sleipner has been so successful:

Property	Value	Significance
Formation depth	800–1,000 m below seabed	Sufficient depth for supercritical CO ₂
Net thickness	200–250 m	Very large vertical storage column
Porosity	30–42%	Exceptionally high — more pore space for CO ₂
Permeability	1,000–3,000 mD	Very high — CO ₂ flows easily; good injectivity
Salinity	~35 g/L	High salinity — water unusable; no competing use
Caprock	Nordland shale, 50–100 m thick	Proven seal; holds natural gas in adjacent fields
Lateral extent	26,000 km ²	Enormous storage capacity beyond Sleipner needs

3. Monitoring: 28 Years of Evidence

Sleipner has been monitored more intensively than any other CCS site in history. The monitoring programme, developed collaboratively by Equinor, the EU, and research institutes including SINTEF and the British Geological Survey, uses multiple complementary technologies:

Technology	What It Detects	Frequency	Key Finding
4D Seismic	CO ₂ plume shape and growth in Utsira formation	Every 2–3 years since 1994	Plume growing upward and spreading laterally as expected; no surprises
Gravimetry	Density changes in reservoir (CO ₂ vs. brine)	Every 5 years	Confirms mass balance consistent with injection volumes
Seafloor monitoring	Gas seeps, anomalies on seabed above storage site	Annual	Zero seafloor anomalies detected; seabed unchanged
Water column sampling	Dissolved CO ₂ in seawater above storage site	Periodic	No CO ₂ anomaly detected in water column

Technology	What It Detects	Frequency	Key Finding
Pressure monitoring	Reservoir pressure during injection	Continuous	Pressure build-up well within modelled expectations
Micro-seismic	Any fault reactivation or induced seismicity	Continuous	No induced seismicity; fault-free behaviour

The Most Important CCS Monitoring Finding of the Last 30 Years: 25+ million tonnes of CO₂ has been injected into the Utsira formation. Zero has leaked. The seabed above Sleipner is geologically indistinguishable from control sites with no injection. This is the empirical proof that properly selected and monitored geological storage works — on timescales that matter.

4. What the Plume Tells Us: CCS Science in Action

The 4D seismic surveys at Sleipner have produced the most detailed picture of subsurface CO₂ behaviour ever obtained. Key scientific findings:

- **Layered migration:** The CO₂ plume has migrated upward through the Utsira formation, accumulating in distinct layers beneath thin intra-formational mudstone lenses. The mudstone layers act as temporary baffles, slowing upward migration and giving more time for trapping mechanisms to activate.
- **Trapping transition:** Over 28 years, the plume has evolved. The initial structural trapping (CO₂ accumulating under caprock) is being augmented by residual trapping (CO₂ droplets in pore spaces), solubility trapping (CO₂ dissolving into brine), and the beginnings of mineral trapping. Each mechanism is more secure than the last.
- **Model calibration:** Comparing actual plume behaviour with numerical models has revealed where the models over- and under-predict CO₂ migration. This calibration work has improved the accuracy of storage site assessments worldwide.

5. Lessons and Relevance for India

- **Geological storage works at scale:** Sleipner's 25 Mt stored over 28 years at ~1 Mt/year demonstrates that geological storage is safe, permanent, and manageable at the scale relevant to industrial decarbonisation.
- **The carbon tax made it happen:** Sleipner would not exist without Norway's offshore CO₂ tax. This is the clearest possible demonstration that carbon pricing drives CCS adoption. India's developing carbon market and potential international carbon finance mechanisms (Article 6) play an equivalent role.
- **India's geological context:** India has several sedimentary basins with potentially suitable storage formations. The Cambay Basin (Gujarat), the Krishna-Godavari Basin (Andhra Pradesh), and offshore formations in the Bay of Bengal and Arabian Sea are areas of active geological investigation. India lacks the 30-year head start of Norway, but the Sleipner methodology is directly applicable.
- **Open science:** Equinor and the EU jointly funded extensive publication of Sleipner monitoring data. This openness has benefited the global CCS community enormously. Any Indian CCS geological storage programme should adopt a similar open-data approach.

Key Takeaway: Sleipner is the proof-of-concept that the world needed. It answers the fundamental question — 'will CO₂ stay underground?' — with 28 years of empirical data. The answer is yes.

Prepared by Meer Foundation EduSuTo | Case Study 2 of 6 | Module 4